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Owner Courtney Doellner: Clinical Pharmacist Specialist– Pediatric Medication
 Area Pharmacy Services
 Applicability UMHS Clinical
 References Guideline

Pharmacy Services IV Guidelines for Michigan Congenital Heart Center (MCHC) Guideline

DRUG – GENERIC NAME [TRADE NAME®]	LOAD/BOLUS DOSING	USUAL DOSING RANGE	NURSING TITRATION INSTRUCTIONS	COMMENTS	ADMINISTRATION CONCERNS
ALPROSTADIL (PGE, PROSTIN, PROSTAGLANDIN) PCTU, MCA	NONE	0.03-0.2 mcg/ kg/min Hard max: 0.4 mcg/kg/min	DO NOT TITRATE		Central line administration - If <1 mL/hr, then D10W (or D5W) WITHOUT heparin should be co-infused - If >1 mL/hr, no fluids needed for co-infusion
AMINOPHYLLINE *IV form of theophylline PCTU	Loading: 5-8 mg/kg IV given over 30 minutes	0.5-1.5 mg/ kg/hr Hard max: 1.5 mg/kg/hr	DO NOT TITRATE	- Follow theophylline levels 1 hour post-load, then Q12-24hr - Therapeutic: 10-15 mcg/mL - Toxicity > 20 mcg/mL	
AMIODARONE PCTU, MCA	Loading: <50kg: 5mg/ kg IV (max 150mg/dose) over 60 minutes May repeat loading dose	<50kg or <18 yos: 5-15 mcg/kg/ min Hard max: 20 mcg/kg/min ≥50kg or ≥18	DO NOT TITRATE		MUST BE FILTERED with 0.22micron filter - Tubing good

	(max 15 mg/kg/ day or 450 mg/day) ≥50kg: 150mg/dose IV over 60 minutes ; may repeat (Max 450 mg/day)	yoa: 1 mg/min x 6hrs, then 0.5 mg/min Hard max: 2 mg/min			for 96 hours - Concentrations greater than 2 mg/mL must go via central line - Risk of phlebitis and extravasation with peripheral use – call Pharmacy (48208) if extravasation occurs Loading dose MUST be given over 60 minutes to avoid hypotension - Diluents: D5W (preferred)
ARGATROBAN PCTU, MCA, 11W	Loading: ONLY used in cath lab or to prime ECMO circuit	0.1-3 mcg/kg/min ECMO: 0.5-5 mcg/kg/min Hard max: 10 mcg/kg/min	DO NOT TITRATE	Follow aPTTs or anti-IIa levels	
AMINOCAPROIC ACID [AMICAR®] PCTU	Loading: 50-100 mg/kg IV (max 5000 mg/dose) given over 15-60 minutes	<50kg: 25-30 mg/kg/hr Hard max: 100 mg/kg/hr Adult order: 1 g/hr Soft max: 1.5 g/hr	DO NOT TITRATE	- Pediatric dosing in mg/kg/hr - Adult dosing in g/hr - Usually reserved for patients on ECMO with bleeding concerns	
BIVALIRUDIN [ANGIOMAX®] PCTU	Loading: 0.15-0.3 mg/kg (not routinely used)	0.1-3 mg/kg/hr Hard max: 5 mg/kg/hr Adult order: 1.75 mg/kg/hr Hard max: 2 mg/kg/hr	DO NOT TITRATE	- Follow aPTTs - Agent of choice for VAD patients	
BUMETANIDE [BUMEX®] PCTU	NONE	0.01-0.05 mg/kg/hr Hard max: 0.08 mg/kg/hr	DO NOT TITRATE		No diluent – this is straight drug
CALCIUM CHLORIDE	NONE	2.5-20 mg/kg/hr	DO NOT TITRATE		CENTRAL LINE

PCTU		Hard max: 30 mg/kg/hr CRRT Calcium Infusion: per CRRT Citrate protocol	Exception: For CRRT calcium infusion, titrate per CRRT Citrate protocol		administration ONLY • Vesicant – if extravasation occurs, call Pharmacy (48208)
CISATRICURIUM [NIMBEX®] PCTU	Bolus: may give 0.1 mg/kg and then initiate continuous infusion	1-10 mcg/kg/min Hard max: 13 mcg/kg/min	DO NOT TITRATE		• The 2 mg/mL has no diluent – this is straight drug
DEXMEDETOMIDINE [PRECEDEX®] PCTU	Due to potential side effects, loading doses are not currently recommended	0.2 – 2 mcg/kg/hr Hard max: 3 mcg/kg/hr	DO NOT TITRATE	• Common side effects: bradycardia and hypotension	
DILTIAZEM [CARDIZEM®] PCTU, MCA	Loading: 0.25 mg/kg IV given over 2 minutes (max 25 mg); after 15 minutes if cardiac rhythm response is not achieved, prescriber may order 2 nd dose of 0.35 mg/kg (max 25 mg)	<50kg: 0.05-0.2 mg/kg/hr Hard max: 0.3 mg/kg/hr ≥50kg: 5-15 mg/hr Hard max: 20 mg/hr	DO NOT TITRATE	• Pediatric dosing in mg/kg/hr • Adult dosing in mg/hr	
DOBUTAMINE PCTU, MCA, 11W	NONE	2.5-20 mcg/kg/min Hard max: 25 mcg/kg/min	Titrate by 2.5-5 mcg/kg/min every 5 minutes until ordered parameter or max ordered dose achieved		• Vesicant – if extravasation occurs, call Pharmacy (48208)
DOPAMINE PCTU, MCA, 11W	NONE	2.5-20 mcg/kg/min Hard max: 25 mcg/kg/min	Titrate by 2.5 mcg/kg/min every 5 minutes until ordered parameter or max ordered dose achieved	• Low dose: 1-5 mcg/kg/min; increase renal blood flow and urine output • Intermediate dose: 5-15 mcg/kg/min; increase heart rate, CO and BP • High dose: >15	• Vesicant – if extravasation occurs, call Pharmacy (48208)

				mcg/kg/min; increase BP, vasoconstriction	
EPINEPHRINE HYDROCHLORIDE PCTU	NONE	0.01-0.1 mcg/ kg/min Hard max: 4 mcg/kg/min	Titrate by 0.01-0.02 mcg/ kg/min every 5 minutes until ordered parameter or max ordered dose achieved		Vesicant – if extravasation occurs, call Pharmacy (48208)
EPOPROSTANOL [FLOLAN®] PCTU, MCA, 11W	NONE	2-20 ng/kg/ min Hard max: 125 ng/kg/ min Starting dose: 2-3 ng/kg/ min; increased based on patient response Typical dose increase by 1-2 ng/kg/min 1 ng (nanogram) = 0.001 mcg	DO NOT TITRATE		Flolan syringes must be changed at least every 72 hours, when the drug syringe expires or when the tubing needs changing per unit policy, which ever comes sooner; protect from light MUST BE FILTERED with 0.22micron filter - Flolan is NOT COMPATIBLE with any other meds or IV fluids; run in dedicated line - DO NOT stop, pause, or interrupt infusion - Diluent: special Flolan diluent
ESMOLOL [BREVIBLOC®] PCTU, MCA	Loading: 50-500mcg/ kg given over 1 minute	50-250 mcg/ kg/min Hard max: 800 mcg/kg/ min	DO NOT TITRATE		No Diluent - straight drug
FENOLDOPAM PCTU	NONE	0.1-0.8 mcg/ kg/min Hard max: 0.8mcg/kg/ min	DO NOT TITRATE		

FENTANYL [SUBLIMAZE®] PCTU	Bolus: 0.5-2 mcg/kg IV given slow IV push	<50kg: 0.5-3 mcg/kg/hr Hard Max: 10 mcg/kg/hr ≥50kg: 25-200 mcg/hr Hard Max: 400 mcg/hr	DO NOT TITRATE	- Pediatric dosing in mcg/kg/hr - Adult dosing in mcg/hr	- Pushing IV fentanyl too quickly may result in chest wall rigidity - The 50 mg/mL concentration has no diluent – straight drug
FUROSEMIDE [LASIX®] PCTU, MCA	NONE	<50kg: 0.3-0.7 mg/kg/hr Hard max: 1.5 mg/kg/hr ≥50kg: 5-40 mg/hr Hard max: 160 mg/hr	DO NOT TITRATE	- Pediatric dosing in mg/kg/hr - Adult dosing in mg/hr	The 10 mg/mL concentration has no diluent – this is straight drug
HEPARIN PCTU, MCA, 11W	Loading: 50-100 units/kg IV given over 15 minutes; per MD request ECMO Loading: 100 units/kg IV x 1	Initial: 10-20 units/kg/hr Hard max: 75 units/kg/hr ECMO: 10-40 units/kg/hr Hard max: 100 units/kg/hr	DO NOT TITRATE	- Monitor aPTT or anti-Xa - Doses will be adjusted per levels	
HYDROMORPHONE	Bolus: usually equal to infusion rate given q1-4h prn	<50kg: 5-20 mcg/kg/hr Hard max: 45 mcg/kg/hr ≥50kg: 0.25-3 mg/hr Hard max: 5 mg/hr	<50kg: titrate by 1 mcg/kg/hr every 4 hours until ordered parameter or max ordered dose achieved >50kg: titrate by 0.1 mg/hr every 4 hours ordered parameter or max ordered dose achieved	- Pediatric dosing in mcg/kg/hr - Adult dosing in mg/hr - Note: hydromorphone is ~7-10x as potent as morphine (i.e. morphine 200 mcg/kg/hr ≈ hydromorphone 20-28 mcg/kg/hr)	Use is restricted for sedation in mechanically ventilated opioid tolerant patients
INSULIN REGULAR PCTU	NONE	Initial: 0.01-0.1 units/kg/hr Hard max: 0.4 units/kg/hr ≥50kg: Initiate at 1-5 units/hr [1 unit/mL concentration ONLY] Hard max: 30	DO NOT TITRATE	- Pediatric dosing in units/kg/hr - Adult dosing in units/hr - For adult-sized patients, refer to the UH Adult Insulin Protocol available on the UM website	MONITOR GLUCOSE LEVELS

ISOPROTERENOL [ISUPREL®] PCTU	NONE	units/hr <50kg: 0.01-2 mcg/kg/min Hard max: 2 mcg/kg/min ≥50kg: 2-10 mcg/min Hard max: 11 mcg/min	DO NOT TITRATE	- Pediatric dosing in mcg/kg/min - Adult dosing in mcg/min	- Concentrations greater than 0.01 mg/mL must go via central line - Vesicant – if extravasation occurs, call Pharmacy (48208)
KETAMINE [KETALAR®] PCTU	Bolus: may give 0.5-2 mg/kg IV over 1-3 minutes	: 5-20 mcg/kg/min Hard max: 80 mcg/kg/min	DO NOT TITRATE	Must use PCTU/PICU profile in Alaris device	
LABETOLOL PCTU, MCA	Bolus: 0.2-1 mg/kg (max 40 mg/dose); may administer over 2 minutes as long as under max of 10 mg/min	<50kg: 0.25-3 mg/kg/hr Hard max: 3 mg/kg/hr ≥50kg: 0.5-2 mg/min Hard max: 6 mg/min	DO NOT TITRATE	- Pediatric dosing in mg/kg/hr - Adult dosing in mg/min	Concentrations greater than 2 mg/mL must go via central line
LEVOTHYROXINE (T4 – GIFT OF LIFE) [SYNTHROID®] PCTU	Loading: 0.8-5 mcg/kg IV; administered over 30 minutes	<50kg: 0.8-1.4 mcg/kg/hr Hard max: 1.4 mcg/kg/hr ≥50kg: 1-20 mcg/hr Hard max: 20 mcg/hr	TITRATE BASED ON ORDER	- Pediatric dosing in mcg/kg/hr - Adult dosing in mcg/hr	Age based dosing available in Lexicomp
LIDOCAINE PCTU, MCA	Loading: 1 mg/kg IV given over 2 minutes followed by continuous IV infusion; may administer second bolus if delay between initial bolus and start of infusion is >15 minutes	<50 kg: 20-50 mcg/kg/min Hard max: 80 mcg/kg/min ≥50kg: 0.5-4 mg/min Hard max: 5 mg/min	DO NOT TITRATE	- Pediatric dosing in mcg/kg/min - Adult dosing in mg/min	- Concentrations greater than 4 mg/mL must go via central line - Check lidocaine levels as requested per MD - Therapeutic: 1.5-5 mcg/mL
MIDAZOLAM [VERSED®] PCTU	Bolus: 0.05-0.2 mg/kg IV given over 2 minutes (max 4 mg/dose)	<50kg: 10-150 mcg/kg/hr Hard max: 300 mcg/kg/hr ≥50kg: 1-10	DO NOT TITRATE	- Pediatric dosing in mcg/kg/hr - Adult dosing in mg/hr	

		mg/hr [1 mg/mL conc ONLY] Hard max: 20 mg/hr			
MILRINONE PCTU, MCA, 11W	NONE	0.3-0.75 mcg/ kg/min Hard max: 1 mcg/kg/min	DO NOT TITRATE	125mg/250mL (0.5 mg/mL) is restricted to continuation of home therapy ONLY	
MORPHINE SULFATE PCTU, MCA*, 11W*	Bolus: 0.1-0.3 mg/kg IV given over 5 minutes	<50kg: 10-150 mcg/kg/hr Hard max: 300 mcg/kg/hr ≥50kg: 1-5 mg/hr Hard max: 20 mg/hr	DO NOT TITRATE	- Pediatric dosing in mcg/kg/hr - Adult dosing in mg/hr	- Monitor for histaminergic/allergic response *MCA/11W concentrations use the 0.05 mg/mL
NICARDIPINE PCTU	NONE	<50 kg: 0.5-5 mcg/kg/min Hard max: 6 mcg/kg/min ≥50kg or ≥18 yoa: 5-15 mg/hr Hard max: 15 mg/hr	DO NOT TITRATE	- Pediatric dosing in mcg/kg/min - Adult dosing in mg/hr	- Concentrations greater than 0.2 mg/mL must go via central line - Change peripheral infusion site every 12 hours to avoid irritation - Risk of phlebitis and extravasation with peripheral use – call Pharmacy (48208) if extravasation occurs
NITROGLYCERIN PCTU	NONE	<50 kg: 0.5-5 mcg/kg/min Hard max: 20 mcg/kg/min ≥50kg or ≥18 yoa: 5-200 mcg/min Hard max: 210 mcg/min	DO NOT TITRATE	- Pediatric dosing in mcg/kg/min - Adult dosing in mcg/min	- Concentrations greater than 0.2 mg/mL must go via central line - Use special IV tubing (low sorbing tubing) available in nursing clean room

NITROPRUSSIDE [NIPRIDE®] PCTU	NONE	0.3-10 mcg/kg/min Hard max: 11 mcg/kg/min Doses > 400 mcg/min not recommended due to increase risk thiocyanate toxicity, with little added benefit	DO NOT TITRATE	- Check cyanide/thiocyanate level every 24 hours to monitor for cyanide toxicity - Cyanide toxicity: >2 mcg/mL - Thiocyanate toxicity: >35 mcg/mL	- Concentrations greater than 0.5 mg/mL must go via central line Protect syringe from light – do not need to cover IV tubing
NOREPINEPHRINE BITARTRATE [LEVOPHED®] PCTU	NONE	0.01-0.1 mcg/kg/min Hard max: 2 mcg/kg/min	Titrate by 0.01-0.02 mcg/kg/min every 5 minutes ordered parameter or max ordered dose achieved		- Concentrations greater than 0.01 mg/mL must go via central line - Vesicant – if extravasation occurs, call Pharmacy (48208)
OCTREOTIDE PCTU, MCA, 11W	Load: for GI bleed, may administer 1-2 mcg/kg IV bolus over 15-30 minutes; followed by continuous infusion No load for chylothorax	Chylothorax: 1-3 mcg/kg/hr Max: 6 mcg/kg/hr GI bleed: 1-10 mcg/kg/hr Hard max: 11 mcg/kg/hr ≥50 kg or ≥18 yoa: 25-50 mcg/hr Hard max: 75 mcg/hr	DO NOT TITRATE	- Pediatric dosing in mcg/kg/hr - Adult dosing in mcg/hr	
PANTOPRAZOLE [PROTONIX®] PCTU, MCA, 11W	Loading: 1-2 mg/kg IV (max 80 mg/dose) given over 5 minutes	<50kg: 0.1-0.2mg/kg/hr Hard max: 0.5 mg/kg/hr ≥50 kg: 5-10 mg/hr Hard max: 12 mg/hr	DO NOT TITRATE	- Pediatric dosing in mg/kg/hr - Adult dosing in mg/hr	Should be used for GI bleeding only
PENTOBARBITAL [NEMBUTAL®] PCTU	Bolus: 1-2 mg/kg IV given over 20 minutes	0.25-7.5 mg/kg/hr Hard max: 15 mg/kg/hr	DO NOT TITRATE		Infusion, tubing, and stopcock needs to be changed Q24hr

					Does NOT need to be filtered
PHENYLEPHRINE PCTU, MCA	Bolus: 5-20 mcg/kg/dose every 10 to 15 minutes as needed	0.05-5 mcg/kg/min Hard max: 5 mcg/kg/min ≥50 kg: 50-300 mcg/min Hard max: 300 mcg/min	Titration available according to physician instructions per MiChart	- Pediatric dosing in mcg/kg/min - Adult dosing in mcg/min	- Vesicant – if extravasation occurs, call Pharmacy (48208) - Central line administration preferred
PROCAINAMIDE PCTU, MCA	Loading: <50kg: 7-15 mg/kg (max 1500 mg/dose) over 30-60 minutes ≥50kg: 10-17 mg/kg at a rate of 20-50 mg/minute or 100 mg every 5 minutes until arrhythmia is controlled, hypotension occurs or QRS complex widens by 50% of its original width	<50kg: 20-80 mcg/kg/min, max of 2000 mg in 24 hr Hard max: 100 mcg/kg/min ≥50kg: 1-4 mg/min Hard max: 6 mg/min	DO NOT TITRATE	- Pediatric dosing in mcg/kg/min - Adult dosing in mg/min - Check Procainamide and NAPA levels 6 to 12 hours after start of infusion per MD - PROC toxicity: >12 mcg/mL - NAPA toxicity: >40 mcg/mL	
PROPOFOL [DIPRIVAN®] PCTU	Loading/Bolus: 0.5-3 mg/kg	10-150 mcg/kg/min Hard max: 150 mcg/kg/min	DO NOT TITRATE	- Restricted to ≤12 hours in patients <18 years old, if using for longer - obtain anesthesia approval - Obtain triglyceride level if continued beyond 24 hours - Do not use in prolonged QT syndrome	Discard infusion AND tubing every 12 hours No diluent – straight drug - Vesicant – if extravasation occurs, call Pharmacy (48208)
ROCURONIUM PCTU	Bolus: 1 mg/kg IV over 5-10 seconds	5-12 mcg/kg/min Hard max: 12 mcg/kg/min	DO NOT TITRATE		Vesicant – if extravasation occurs, call Pharmacy

					(48208)
SILDENAFIL PCTU	NONE	Typical starting dose: 0.067 mg/kg/hr (= 1.67 mg/kg/DAY) Dosing range: 0.015-0.25 mg/kg/hr Hard max: 1.67 mg/hr	DO NOT TITRATE		Sildenafil is compatible with D5W but MUST run in dedicated line
SOTALOL PCTU	Loading: 1 mg/kg IV given over 1 hour followed by continuous IV infusion	4.5 mg/kg/DAY Hard max: 12 mg/kg/DAY	DO NOT TITRATE		IV sotalol is compatible with D5W and NS but MUST run in a dedicated line
TERBUTALINE PCTU	Loading: 2-10mcg/kg IV over 5 minutes	0.08-4 mcg/kg/min Hard max: 4 mcg/kg/min	DO NOT TITRATE		
TREPROSTINIL (REMODULIN®) PCTU, MCA, 11W	NONE	0.625-40 ng/kg/min Starting dose: 0.625-1.25 ng/kg/min; increased based on patient response Typical dose increase by 1.25-2.5 ng/kg/min 1 ng (nanogram) = 0.001 mcg	DO NOT TITRATE		Remodulin IV syringes/cassette must be changed every 48 hours IV Tubing must be changed every 48 hours MUST BE FILTERED with 0.22micron filter - Administer through a dedicated line - Diluted treprostinil is stable for 48 hours - Avoid abrupt withdrawal
VASOPRESSIN (SHOCK) PCTU	NONE	<50kg: 0.0001-0.003 unit/kg/min Hard max: 0.003 unit/kg/min ≥50kg: 1-5 units/hr	DO NOT TITRATE	- Pediatric dosing in units/kg/min - Adult dosing in units/hr NOTE: shock dosing and DI dosing are	- Concentrations greater than 0.4 units/mL must go via central line - Vesicant – if extravasation

		Hard max: 5 units/hr		DIFFERENT	occurs, call Pharmacy (48208)
VASOPRESSIN (DIABETES INSIPIDUS) PCTU	NONE	0.25-10 milliunit/kg/hr Hard max: 40 milliunit/kg/hr	DO NOT TITRATE	• NOTE: shock dosing and DI dosing are DIFFERENT • Dosage is highly variable and is titrated based on serum and urine sodium and osmolality in addition to fluid balance and urine output	• Vesicant – if extravasation occurs, call Pharmacy (48208)
VECURONIUM PCTU	Bolus: 0.1 mg/kg IV over 5-10 seconds	0.5-2 mcg/kg/min Hard max: 4 mcg/kg/min	DO NOT TITRATE		• Vesicant – if extravasation occurs, call Pharmacy (48208) • No diluent – straight drug

***Note: adult dosing refers to patients >50 kg**

***Doses listed above are suggested maximum doses for patient safety; start at lowest dose when titrating unless clinical condition requires higher dose. Some clinical conditions may warrant the use of doses higher than those listed above.**

Approved:

Pediatric Medication Safety: 2/2013, 2/2019, 12/2019, 8/2020, 4/2021

Pharmacy & Therapeutics Committee: 2/2013, 4/2019, 1/2020, 9/2020, 5/2021

Approval Signatures

Step Description

Approver

Date